


January 8, 2026

Base 2, 8, 16

$$\begin{array}{cccccccccccc} & 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 & & & & & \\ \text{Binary } (11111111) & \rightarrow & (& & & & & &) & & & & & \\ & & & & & & & & & & & & & \end{array}$$
$$1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$
$$= 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 =$$
$$= 255_{10}$$

$$(377)_8 = 3 \cdot 8^2 + 7 \cdot 8^1 + 7 \cdot 8^0 = 255_{10}$$

$$(FF)_{16} \rightarrow ()_{10}$$

base 16:

$$16^{\overset{1}{F}} + 16^{\circ F} = 16 \cdot 15 + 15 = 255_{10}$$

Decimal
 $()_{10} \rightarrow ()_2$
 $()_{10} \rightarrow ()_8$
 $()_{10} \rightarrow ()_{16}$

10	A
11	B
12	C
13	D
14	E
15	F

HEX

255		2						
254		127		2				
		126		63		2		
				62		31		2
						30		15
								7
								6
								3
								2
								1

255 = 11111111

10

2

1

1

1

1

1

114₁₀

$$\begin{array}{r|l} 114 & 2 \\ \hline 114 & 57 \\ \hline 0 & 56 \\ & 28 \\ & 14 \\ & 7 \\ & 3 \\ & 1 \end{array}$$

1110010₂

$$(0.12)_{10} \rightarrow ()_2 = (.000111)_2$$

$0.12 \times 2 = 0.24$	0	just fraction ↓
$0.24 \times 2 = 0.48$	0	
$0.48 \times 2 = 0.96$	0	
$0.96 \times 2 = 1.92$	1	
$0.92 \times 2 = 1.84$	1	
$0.84 \times 2 = 1.68$	1	

Dec.	Binary	Dec.	Binary
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111
8	1000	hex.	

Binary addition

$$\begin{array}{r}
 \times \quad 01100 \\
 \quad 10001 \\
 \hline
 11101
 \end{array}$$

$$\begin{array}{r}
 \overset{1}{+} \overset{1}{1} \overset{1}{0} \overset{1}{1} 1 0 \\
 \underline{10111} \\
 1[01101]
 \end{array}$$

$$(1)_2 + (1)_2 = (10)_2$$

$$(10)_2 + (1)_2 = (11)_2$$

It is overflow, machine should know that it is overflow

Binary subtraction

$$\begin{array}{r}
 \overset{0}{-} \overset{0}{0} \\
 \hline
 0
 \end{array}
 \quad
 \begin{array}{r}
 \overset{1}{-} \overset{1}{1} \\
 \hline
 0
 \end{array}
 \quad
 \begin{array}{r}
 \overset{1}{-} \overset{0}{0} \\
 \hline
 1
 \end{array}
 \quad
 \begin{array}{r}
 \overset{0}{-} \overset{1}{1} \\
 \hline
 1
 \end{array}$$

$$\begin{array}{r}
 -10110 \\
 \underline{10010} \\
 00100
 \end{array}$$

$$\begin{array}{r}
 -10110 \\
 \underline{10011} \\
 00011
 \end{array}$$

1's complement

logic operation

$$(1010111)_2$$

1's complement is inverse of digits

$$(01010000)_2 \text{ 1's comp.}$$

$$(01010001)_2 \text{ 2's complement}$$

$$(0101)_2 = (5)_{10}$$

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$(r-1)$'s r 's complements

radix = base

$$r=10 \quad L=4 \quad N=2468 \quad r^4=10000$$

$$r^4 - 1 = 9999$$

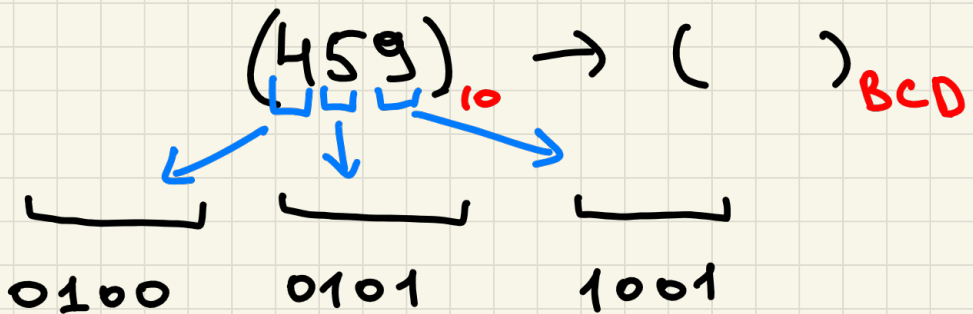
Therefore

$$\begin{array}{r} 9999 \\ - 2468 \\ \hline \underline{7531} \end{array}$$

Decimal $(185)_{10}$

Binary $(10111001)_2$

Binary Coded Decimal BCD



$$4_{10} + 8_{10} = 12_{10}$$

BCD range
(0-9)

$$0100_2 + 1000_2 = 1100_2 \rightarrow 12_{10}$$

$$0110_2 \rightarrow 6_{10}$$

$(\underline{0001} \underline{0010})_{BCD}$

$$\begin{array}{r}
 + 162 \\
 769 \\
 \hline
 931
 \end{array}$$

Binary Coded Decimal

$$\begin{array}{r}
 0001 \quad 0110 \quad 0010 \\
 0111 \quad 0110 \quad 1001 \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 1000 \quad \underline{1100} \quad \underline{1011} \\
 \text{doesn't exist in BCD} \quad \text{more than 9} \\
 \text{to solve add } 6_{10}
 \end{array}$$

$$\begin{array}{r}
 0110 \quad 0110 \\
 \hline
 \underline{1001} \quad \underline{0011} \quad \underline{0001} \\
 9 \quad 3 \quad 1
 \end{array}$$

encoding $\rightarrow$ public UTF-8
 encryption $\rightarrow$ special key

BOOLEAN LOGIC

$$A+B \rightarrow A \text{ or } B$$

$$A.B \rightarrow A \text{ and } B$$

$$A', \bar{A} \rightarrow \text{NOT } A$$

inputs

A	B	A.B	A+B	A'	B'
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	0	1
1	1	1	1	0	0

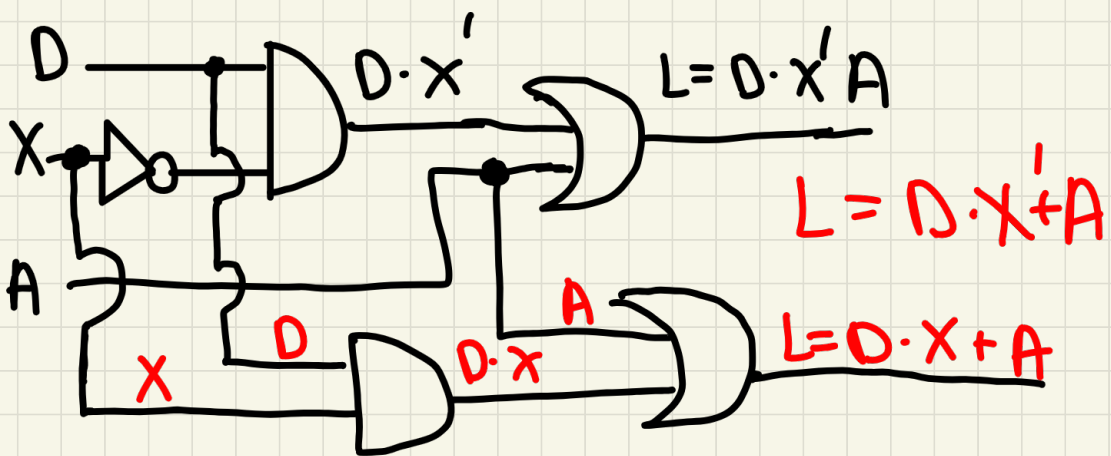
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$$L = \underbrace{D \cdot X'}_{\text{inputs}} + \underbrace{A}_{\text{output}}$$

D	X	A	X'	D · X'	D · X' + A
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0

output

0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1



A	B	C	AB	A'C	BC		AB + A'C
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	0	0	0	0
0	1	1	1	0	1	1	1
1	0	0	0	0	0	0	0

1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	0	1	1	1

$$F(A, B, C) = (A+B)(A+C)$$

$$AA + AC + AB + BC \quad \text{Distributivity}$$

$$A + AC + AB + BC \quad A = AA$$

$$A(\underbrace{1 + C + B}_1) + BC$$

$$A \cdot 1 + BC = A + BC$$

Sum of Products SOP

$$F(A, B, C) = A'B'C + ABC + A'BC'$$

Product of Sum

$$\begin{aligned} F(A, B, C) &= (A+B)(A+C) \\ &= \underbrace{AA + AC + AB + BC}_{\substack{\downarrow \\ A + B = A \\ A + C = A}} \\ &= A(1 + C + B) + BC \\ &= A \cdot 1 + BC = A + BC \end{aligned}$$

$$G = (A + B) \cdot (C + D + E)$$

$$= \underbrace{A \cdot C}_{\text{Product of Sum terms}} + \underbrace{A \cdot D}_{\text{Product of Sum terms}} + \underbrace{A \cdot E}_{\text{Product of Sum terms}} + \underbrace{B \cdot C}_{\text{Product of Sum terms}} + \underbrace{B \cdot D}_{\text{Product of Sum terms}} + \underbrace{B \cdot E}_{\text{Product of Sum terms}}$$

(k)	D _k	X _k	A _k	X'	D _k X	D _k X + A _k	(S _k)	1	X
0	0	0	0	1	0	0	← (D ⁰ X'A)	0	X'
1	0	0	1	0	0	0			
2	0	0	1	0	0	0			
3	0	1	1	0	0	1	← D ³ X'A		
4	1	0	0	1	1	1			
5	1	0	1	1	1	1	← D ⁵ X'A		
6	1	1	0	0	0	0			
7	1	1	1	0	0	1	← D ⁷ A		

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Q 1

$$y + x'z + xy' = x + y + z$$

$$\underline{y + xy'} + x'z$$

$$(y+x)(y+y') + x'z$$

$$y + x + x'z$$

$$y + (x + x')(x + z)$$

$$y + x + z = x + y + z$$

Q₂

$$x'y' + x'y + xy = x'y$$

$$x'(y' + y) + xy$$

$$x' + xy$$

$$(x' + x)(x' + y)$$

$$1 \cdot x' + y = x'y$$

Q₃

$$x'y' + y'z + xz + xy + yz' \quad 13 \text{ operations}$$

$$x'y' + y'z(x + x') + xz + xy + yz'$$

$$\underline{x'y'} + xy'z + \underline{x'y'}z + xz + xy + yz'$$

$$x'y'(1 + z) + xy'z + xz + xy + yz'$$

$$x'y' + \underline{xy'z} + \underline{xz} + xy + yz'$$

$$x'y' + xz(y' + 1) + xy + yz'$$

$$x'y' + xz + xy + yz'$$

$$x'y' + xz + xy(z+z') + yz'$$

$$x'y' + xz + xyz + xyz' + yz'$$

$$x'y' + xz + xyz + yz'(x+1)$$

$$x'y' + \underline{xz} + \underline{xyz} + yz'$$

$$x'y' + xz(1+z) + yz'$$

$$x'y' + xz + yz'$$

8 operations

SOP

$$F(a,b,c) = a + bc + abc'$$

$$A(a,b,c) = \underbrace{abc}_{\text{Minterms}} + \underbrace{a'bc}_{\text{Minterms}} + \underbrace{abc'}_{\text{Minterms}} \quad \text{Canonical (product terms)}$$

K						Minterms
	a	b	c	A	A	SOP
0	0	0	0	$a'b'c'$	0	$x=1$ $x'=0$
1	0	0	1	$a'b'c$	0	
2	0	1	0	$a'bc'$	0	
3	0	1	1	$a'bc$	1	
4	1	0	0	$ab'c'$	0	
5	1	0	1	$ab'c$	0	
6	1	1	0	abc'	1	
7	1	1	1	abc	1	

$$A(a,b,c) = \sum m(3,6,7) \rightarrow \text{minterms}$$

↓
sum of products

POS (product of sum)

$$G(a,b,c) = \underline{(a+b+c')} \cdot \underline{(a'b'+c')} \cdot \underline{(a+b+c)}$$

Max terms:

$$x=0$$

POS

$$x'=1$$

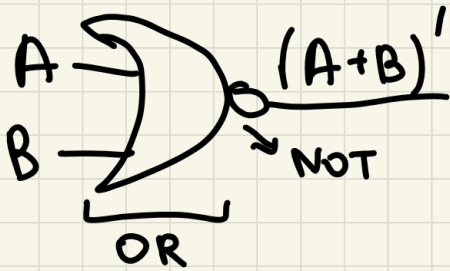
	a	b	c	P	
0	0	0	0	abc	1
1	0	0	1	abc'	1
2	0	1	0	$ab'c$	0
3	0	1	1	$ab'c'$	0
4	1	0	0	$a'b c$	0
5	1	0	1	$a'b c'$	0
6	1	1	0	$a'b'c$	0
7	1	1	1	$a'b'c'$	1

$$P(a,b,c) = \underbrace{\Pi M}_{\text{POS}} (\underbrace{0, 1, 7}_{\text{max terms}})$$

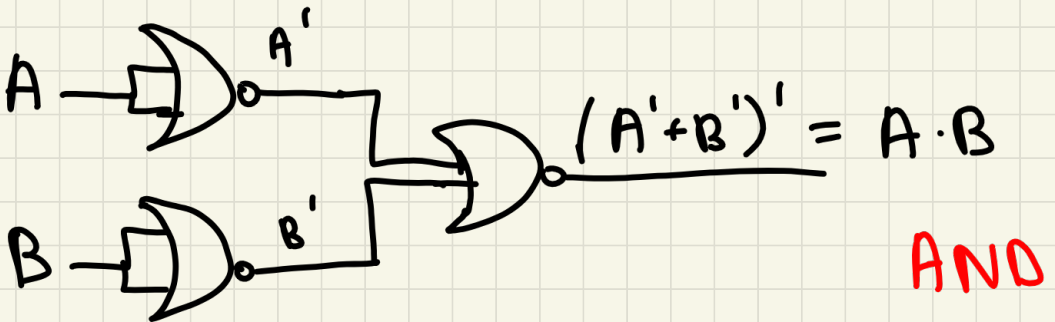
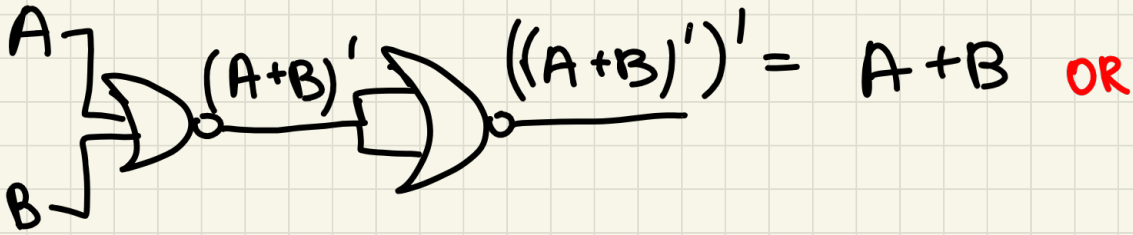
Always write Maxterms POS $x'=1$
Minterms SOP $x=1$
 $x'=0$

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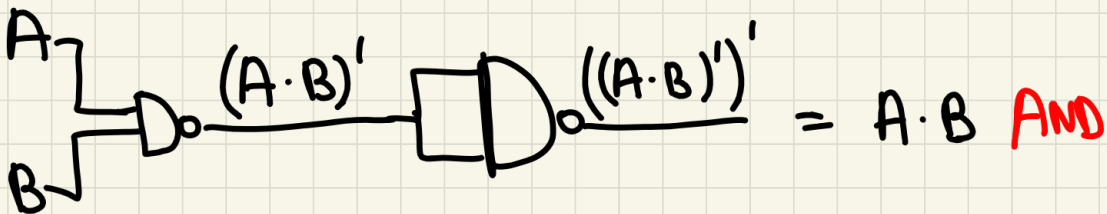
NOR Gate - Universal Gate means
Function completeness (OR, AND, NOT)



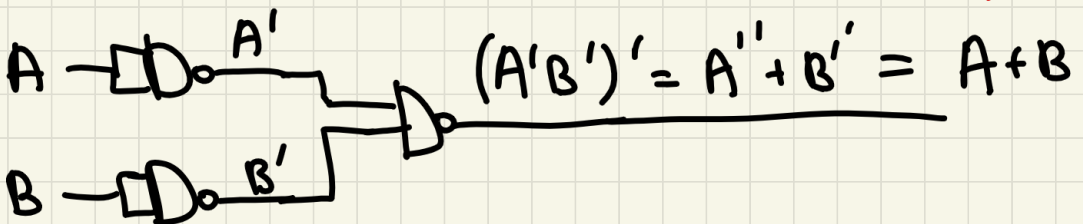
NOR



AND



OR



$$F(a, b, c) = a'b'c + bc' + ac' \text{ (Standard)}$$

↓
missing a
↓
missing b

$$F(a, b, c) = a'b'c + \cancel{abc'} + a'bc' + \cancel{abc'} + \underline{ab'c'}$$

$$F(a, b, c) = a'b'c + a'bc' + abc' + ab'c'$$

SOP Canonical form

$$F(a, b, c) = (a+b+c) \cdot (a+b) \cdot (a+c') \text{ (Standard)}$$

↓
needs c
↓
needs B

$$\cancel{(a+b+c)} \cdot \underline{(a+b+c)} \cdot \cancel{(a+b+c')} \cdot \underline{(a+b+c')} \cdot (a+b'+c')$$

$$(a+b+c) (a+b+c') (a+b'+c')$$

POS Canonical form

1-Variable map

x	F(x)
0	F(0)
1	F(1)

2 variable maps

x	y	F(x)
0	0	F(0)
0	1	F(1)
1	0	F(2)
1	1	F(3)

	0	1
0	00	01
1	10	11

2 variable maps

x	y	z	F(x)
0	0	0	F(0)
0	0	1	F(1)
0	1	0	F(2)
0	1	1	F(3)
1	0	0	F(4)
1	0	1	F(5)
1	1	0	F(6)
1	1	1	F(7)

x \ yz	00	01	11	10
0	$x'z'y'$ 000 0	$x'y'z$ 001 1	$x'yz$ 011 3	$x'yz'$ 010 2
1	xyz 100 4	$xy'z$ 101 5	$x'yz$ 111 7	$x'yz'$ 110 6

→ gray codes

SOP | $x=1$
 $x'=0$

$$F(x,y,z) = \sum m(1,3,5,7)$$

$$x'y'(z+z')$$

$$x'y'(1) = x'y'$$

0	0 ₀	1 ₁	1 ₃	0 ₂
1	0 ₄	1 ₅	1 ₇	0 ₆

Group of 4

$$(2^n) = 1, 2, 4, 8, 16$$

$$x' y' z + x' y z + x y z' + x y z$$

$$z (x' y' + x' y + x y' + x y)$$

$$z (x' \underbrace{(y' + y)}_1 + x \underbrace{(y' + y)}_1)$$

$$z (\underbrace{x' + x}_1)$$

z

Shorter way to calculate

4 variable map

w x y z | f(w, x, y, z)

16 rows

w \ x \ yz	00	01	11	10
00	0000 0	0001 1	0011 3	0010 2
01	0100 4	0101 5	0111 7	0110 6
11	1100 12	1101 13	1111 15	1110 14
10	1000 8	1001 9	1011 11	1010 10

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$$F(a, b, c) = \underbrace{a \cdot b \cdot c'}_{\text{and}} + \underbrace{ab}_{\text{and}} + c$$

OR OR

x

	1	1	
	1	1	

$$SOP = x'y + z$$

② implicant

a

	bc	$b'c'$	$b'c$	bc'
a'	00	01	11	10
a'	0	0	1	1
a	1	0	1	0

$ab'c'$

$a'c$

bc

$$F(a, b, c) = \sum m(1, 3, 4, 7)$$

$$F(a, b, c) = ab'c' + a'c + bc$$

a

	bc	b'c'	b'c	bc'
a'	1	1	1	1
a	0	1	1	0

$$a'(b'c' + b'c + bc + bc') = a'$$

$$a' + c$$

wx \ yz		yz			
		y'z'	y'z	yz	yz'
wx	w'x'	1 ₀	0 ₁	0 ₃	1 ₂
	w'x	1 ₄	1 ₅ *	0 ₇	0 ₆
	wx	1 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
	wx'	1 ₈	0 ₉	1 ₁₀	1 ₁₁

Grouping annotations: Blue circles around (0,0), (0,1), (1,0), (1,1) and (1,2), (1,3), (2,0), (2,1). Green circles around (1,1), (1,2), (2,0), (2,1). Purple arrows pointing to (0,1), (1,1), (2,0), (2,1), (2,2) with labels 1, 4, 5, 3 respectively.

Grouping:

groupsize 8, 4, 2, 1

$$F(w, x, y, z) = (0, 2, 4, 5, 8, 10, 11, 12, 13, 15)$$

$$F(w, x, y, z) = 1, 2, 3, 4, 5$$

1	0	0	1
1	1	0	0
1	1	1	0
1	0	1	1

Red grouping boxes: (0,0), (0,1), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2), (2,3).

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$$F(A, B, C, D) = \sum m(1, 3, 4, 5, 9, 10, 11, 12, 13, 14, 15)$$

AB \ CD				
	00	01	11	10
00		1	1	
01	1	1		
11	1	1	1	1
10		1	1	1

$$F(A, B, C, D) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$$

AB \ CD				
	00	01	11	10
00		1		
01		1	1	1
11	1	1	1	
10			1	

Prime : $BD, A'C'D, A'BC, ABc', ACD$

Combinational Logic Gates

Half adder

$$\begin{array}{r} A \\ + B \\ \hline \text{Sum} \end{array}$$

2 inputs : A, B

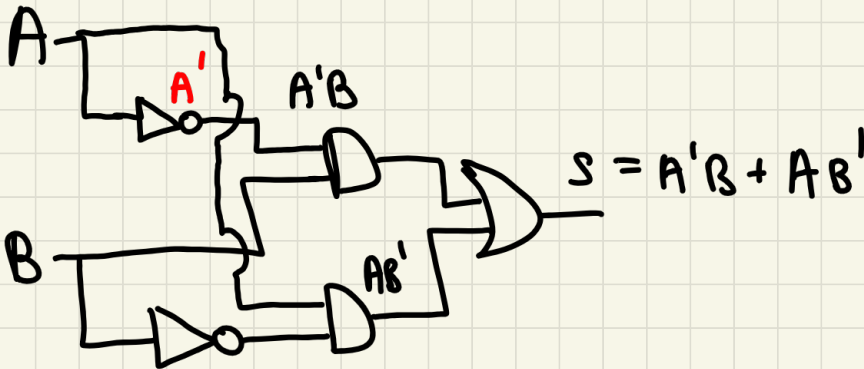
2 outputs : S, C

inputs
↔

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$S = A'B + AB'$$

$$C = AB$$



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Full adder

A	B	C_{in}	S_{out}	C_{carry}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

S_{out}

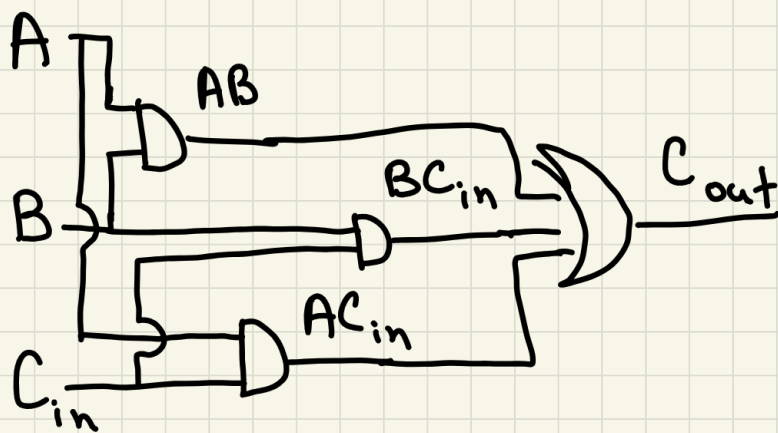
A \ BC				
	00	01	11	10
00	0 ₀	1 ₁	0 ₃	1 ₂
01	1 ₄	0 ₅	1 ₇	0 ₆

$$S = AB'C'_{in} + A'B'C_{in} + ABC + A'BC'_{in}$$

C_{out}

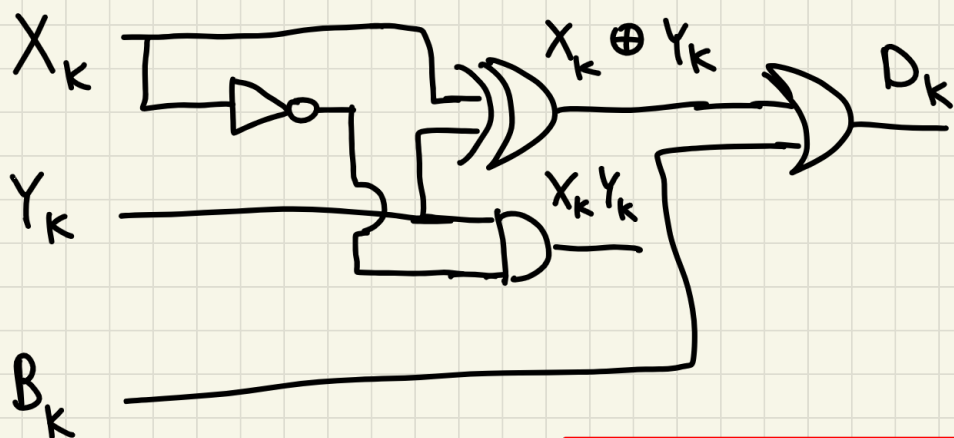
A	BC			
	00	01	11	10
00	0 ₀	0 ₁	1 ₃	0 ₂
01	0 ₄	1 ₅	1 ₇	1 ₆

$$C_{out} = AB + AC_{in} + BC_{in}$$



Inputs	X	Y	B_k	D	B_{k+1}
	0	0	0	0	0
	0	0	1	0	
	0	1	0	0	
	0	1	1	0	
	1	0	0	0	
	1	0	1	0	
	1	1	0	0	
	1	1	1	0	

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A_1	A_0	B_1	B_0	E	G	L
0	0	0	0	1	0	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	1	0
1	0	0	1	0	1	0
1	0	1	0	1	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	0
1	1	1	0	0	1	0
1	1	1	1	1	0	0

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Decoders

2 input		4 output				$2^2=4$
A_1	A_0	D_0	D_1	D_2	D_3	
0	0	1				
0	1		1			
1	0			1		
1	1				1	

diagonal 1
rest 0

A_2, A_1, A_0 | $D_0, D_1, D_2, D_3, D_4, D_5, D_6, D_7$

0
1
2
3
4
5
6

1

1

1

1

BCD

	X_3	X_2	X_1	X_0	Z_0	Z_1	Z_2	Z_3	Z_4	Z_5	Z_6	Z_7
0	0	0	0	0	1	1	1	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	1	0	0
2	0	0	1	0	1	0	1	1	1	0	1	0
3	0	0	1	1	1	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1	0	1	1	0
5	0	1	0	1	0	1	1	1	0	1	0	0
6	0	1	1	0								
7	0	1	1	1								
8	1	0	0	0								
9	1	0	0	1								

(0-9)

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

